

Jacob B. Chugg

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EDUCATION

Brigham Young University – Ira A. Fulton College of Engineering

Provo, UT

Bachelor of Science in Mechanical Engineering, Expected Apr 2028

- GPA 3.69 / 4.00 | Honors: 4 merit scholarships covering full tuition + stipend, incl. Charles R. Bluemel Endowed | ASME Member; Leadership Certificate Program, Weidman Center

SKILLS

- **CAD & Design:** SolidWorks (CSWA), GD&T fundamentals, parametric modeling, engineering drawings, BOMs
- **Manufacturing:** Manual machining, turning, milling, sand casting, tolerancing, DFM, assembly & testing
- **Programming & Tools:** Python, C/C++, Arduino, Excel | **Languages:** Tagalog (fluent)

EXPERIENCE

Brigham Young University – Mechanical Engineering Department

Provo, UT

Research Assistant – Combustion Chamber

Apr 2026 – Present

- Designed a parametric combustion chamber assembly in SolidWorks from heat-transfer-based refractory requirements
- Iterated from rectangular to cylindrical geometry to improve manufacturability with Valar Atomics
- Created toleranced drawings, BOMs, and exploded views for a 14-part system with 3 subassemblies; coordinated heat-exchanger interfaces

Brigham Young University – Math Department

Provo, UT

Calculus Teaching Assistant

Aug 2025 – Present

- Instruct hundreds of students in Calculus I and II through individual tutoring and exam review sessions
- Mentor and train new calculus tutors on instructional techniques, problem-solving approaches, and effective student support

Provo Missionary Training Center (MTC)

Provo, UT

Tagalog Language Instructor

Jul 2024 – Jan 2025

- Taught 4 districts (9 – 12 missionaries each) through full-immersion Tagalog instruction, adapting curriculum to district needs to build comprehension and lesson-teaching fluency within 6–12 weeks

PROJECTS

MOI-Optimized Putter Design & Manufacturing

Provo, UT

Personal Project

Jun 2026 – Sep 2026

- Designed and iterated through 3 mallet-style putter head concepts in SolidWorks using mass-property and MOI analysis, balancing mass distribution to align the shaft axis through the center of mass
- Manufactured the putter with a CNC-machined 6061 aluminum head and manually turned brass weights
- Assembled and tested the completed putter for face stability, alignment, and roll consistency

Pipsqueak Engine Design & Manufacturing

Provo, UT

Brigham Young University

Apr 2026 – Aug 2026

- Modeled, machined, assembled, and tested a functional Pipsqueak engine from SolidWorks drawings, applying manual turning, milling, sand casting, uniaxial powder pressing, tolerancing, fits, and DFM principles

Autonomous Sensor-Based Robot

Provo, UT

Brigham Young University

May 2026 – Jun 2026

- Designed, integrated, and tested a battery-powered Arduino robot with photodiode steering, servo tilt-tracking, sonar collision detection, and DC motor control
- Troubleshoot sensor, motor, and embedded-control integration to achieve reliable autonomous operation

LEADERSHIP & SERVICE

The Church of Jesus Christ of Latter-day Saints

Legazpi, Philippines

Volunteer Missionary | Traveling Assistant

Jun 2022 – Jun 2024

- Directly coached zone and district leaders, equipping them to improve teaching and goal-setting across 200+ missionaries
- Communicated daily in Tagalog, building relationships across diverse communities and adapting teaching to individual needs

AWARDS, ACTIVITIES & INTERESTS

- *Koch Family Foundation Scholarship*, University of Utah (2021–2022) | *Centrifugal Pump Competition* – achieved highest flow rate (2021) | *Walk-On Cornerback*, Utah Football (2021) | *Eagle Scout* (2019) | Interests: Golf, Weightlifting